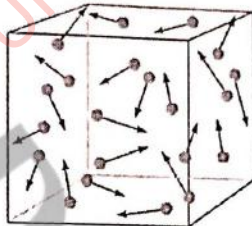


CHAPTER#04**"GASES"****DEFINITION:**

"All those substances neither which possess definite shape nor definite volume are known as gases."

**"BEHAVIOURS"**

Some of the properties of gases are given below:

"DIFFUSIBILITY"**DEFINITION:**

"The entering of particles of one Gas into the spaces between the Particles of other gases is called Diffusion."

KMT EXPLANATION:

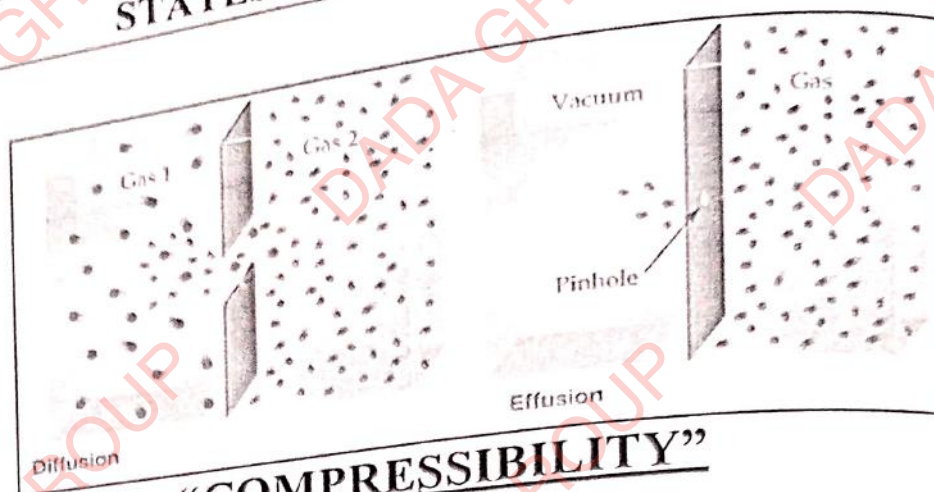
A/c to K.M.T, "The gases molecules are present at large distance i.e. there are large empty spaces in the gases. They do not have any attractive or repulsive forces. They are in the state of continuous random motion."

EXAMPLE:

✓ Diffusion of perfume in air.

EFFUSION:

"It is the process by which a gas escapes from the container through a very small hole. It's the opposite process to diffusion."



"COMPRESSIBILITY"

DEFINITION:

"All the gases are highly compressible. That is, on applying high Pressure, the volume of gas is reduced and the gas is compressed."

KMT EXPLANATION:

According to K.M.T, "The molecules of gas have very large empty spaces between them. When pressure is applied, the molecules come closer and thus the gas is compressed."

Example:

- ✓ The pumping of air into a football or a tire.

"PRESSURE"

DEFINITION:

"The pressure is defined as the force exerted by the gas molecules per unit area."

$$P = \frac{F}{A}$$

K.M.T EXPLANATION:

The molecules of gases are in a state of continuous random motion. These molecules collide with one another and with the wall of the container. The pressure of the gas is due to the collision of gas molecules with the walls. Pressure of a gas can be measured in several units whose relation is given by

$$14.7 \text{ psi} = 760 \text{ mm of Hg} = 760 \text{ torr} = 76 \text{ cm of Hg} = 1 \text{ atm}$$

- The mass of atmosphere on our body at sea level and at 0°C is 14.7 psi or there is about 20 tons total pressure on our bodies.
- The tires of automobiles are filled with air until the gauge shows the pressure of about 28 psi. this means that the pressure inside the tire is 28 psi, greater than the outside pressure; since the external pressure is 14.7 psi, hence the total pressure inside the tire is $28 + 14.7 = 42.7$ psi.

"KINETIC MOLECULAR THEORY"

DEFINITION:

"The internal structure of gases, which explains the behavior and also their behavior towards the gas laws, is called the kinetic molecular theory."

POSTULATES:

Following are the postulates of kinetic molecular theory of gases.

✓ COMPOSITION:

The gaseous state is the one in which the molecules are widely separated from one another having negligible volume.

✓ INTERMOLECULAR SPACES:

These gases are easy to compress due to large empty spaces between the molecules.

✓ MOVEMENT:

The gas particles are in continuous motion, travelling in straight paths between the collisions but in random direction.

✓ MEAN FREE PATH:

The distance between two consecutive collisions of molecules is called free path. The average distance which a molecule travels before its collision is called mean free path.

✓ ELASTIC COLLISION:

The molecules of gas collide with one another and with the walls of container but these collisions are perfectly elastic.

✓ PRESSURE:

The gas pressure is the result of collision of gas molecules with the walls of container.

$$P = F/A$$

✓ INTERMOLECULAR FORCES:

In an ideal gas there are no attractive or repulsive forces between the molecules thus each molecule acts quite independently of the other.

✓ KINETIC ENERGY:

The average K.E of gas molecules depend upon absolute temperature.

$$K.E \propto \text{Temperature}$$

CONCLUSION:

On the basis of above postulates R.J Clausius derived a kinetic equation.

$$PV = \frac{1}{3}mn\bar{c}^2$$

Where,

P = pressure

V = volume

m = mass of one molecule of the gas

n = number of moles of the gas molecules

$\bar{c}^2$ = mean square velocity

"GAS LAWS"

DEFINITION:

"The laws which give the relationship between pressure, volume, temperature and m gases are called gas laws."

Following are the important gas laws:

- ✓ Boyle's Law
- ✓ Charles's Law
- ✓ Avogadro's Law
- ✓ Graham's Law of Diffusion
- ✓ Dalton's Law of partial pressure

"BOYLE'S LAW"

INTRODUCTION:

YEAR: 1660

SCIENTIST: Robert Boyle

STATEMENT:

"The volume of a given mass of gas is inversely proportional to the pressure exerted given temperature."

MATHEMATICALLY:

Let 'V' be the volume of given mass of a gas and 'P' be its pressure, then according to the

Boyle's law

$$V \propto 1/P \quad (\text{at constant temp})$$

$$V = (\text{constant}) 1/P$$

$$V = K/P$$

$$PV = K \quad (\text{where } k \text{ is constant of proportionality})$$

It means that at constant temperature, the product of volume and pressure of a certain mass of a gas is always constant.

FOR INITIAL STATE OF A GAS:

$$P_1 V_1 = K \quad \text{----- 1}$$

FOR FINAL STATE OF A GAS:

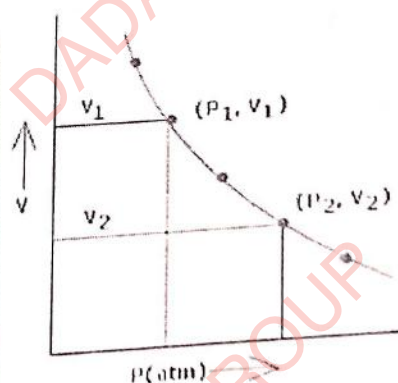
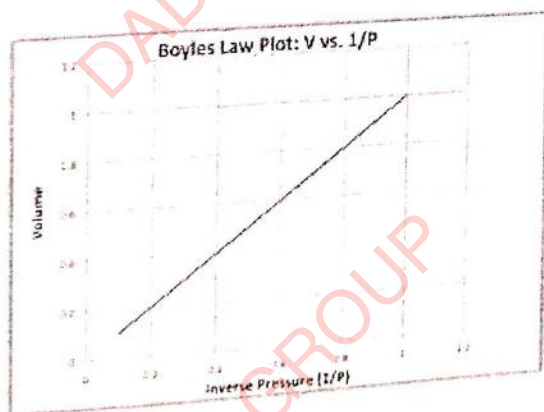
$$P_2 V_2 = K \quad \text{----- 2}$$

Comparing eq. 1 and eq. 2, we get:

$$P_1 V_1 = P_2 V_2$$

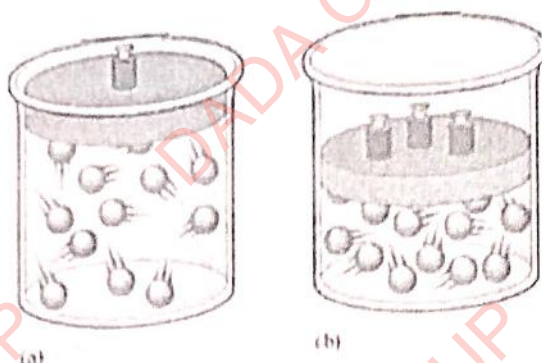
GRAPHICALLY:

The graph between volume and pressure at constant temperature is a parabolic curve and the graph between V and 1/P is a straight line.

**K.M.T EXPLANATION:**

According to K.M.T, "the gaseous state is one in which gas molecules are widely separated from one another.

When pressure of gas is increased, the molecules come closer and as a result volume decreases. When volume is decreased, the molecules now collide more frequently with the walls of container. Hence pressure is increased.

**LIMITATIONS OF BOYLE'S LAW:**

- ✓ Boyle's law is applicable at constant temperature.
- ✓ Boyle's law is not applicable at high pressure and low temperature.

"CHARLE'S LAW"**INTRODUCTION:**

YEAR: 1787

SCIENTIST: French physicist Charles.

STATEMENT:

"The volume of given mass of a gas is directly proportional to the absolute temperature at a constant temperature."

MATHEMATICALLY:

Let "V" be the volume of given mass of a gas and 'T' be its absolute temperature then according to Charles's law,

$$V \propto T \quad (\text{At constant pressure})$$

$$V = (\text{constant}) T$$

$$V/T = \text{constant}$$

$$V/T = K$$

Where "K" is the constant of proportionality.

It means that at constant pressure, the ratio of volume and temperature of a certain mass of a gas is always constant.

For initial stage:

$$V_1/T_1 = K \quad \text{----- (i)}$$

For final stage:

$$V_2/T_2 = K$$

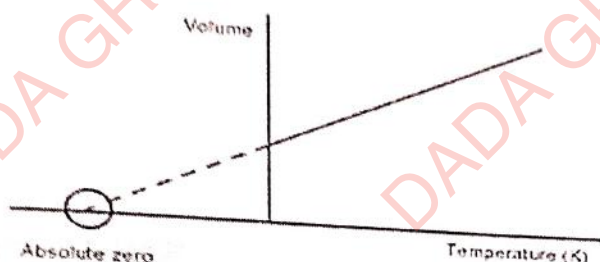
----- (ii)

By comparing equations (i) and (ii),

$$V_1/T_1 = V_2/T_2$$

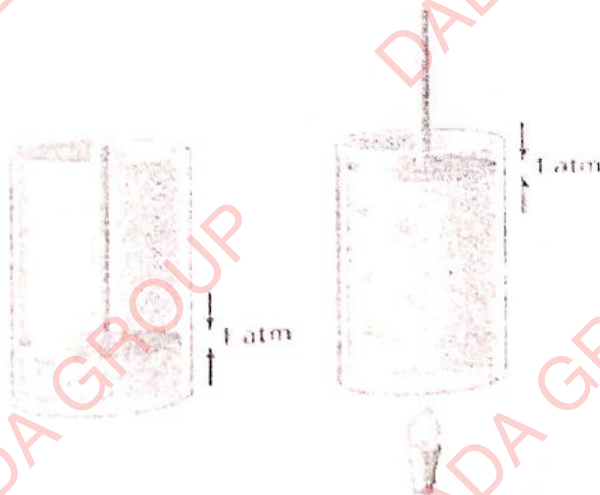
GRAPHICALLY:

If the graph is plotted between V and T then few points can be obtained to draw the straight line. But we cannot obtain points up to -273.16°C . This is theoretically possible but practically it is impossible.



K.M.T EXPLANATION:

According to kinetic molecular theory the gas molecules possess average kinetic energy which is directly proportional to the absolute temperature. By increasing the temperature, the K.E of the gas molecules increase which increase the rate of collisions the molecules with the walls of the container i.e. The pressure is increased. This moves the piston in the upward direction till the rate of collisions become equal to the original value. Hence the volume of the gas is increased.



LIMITATIONS OF CHARLE'S LAW:

- ✓ Charles's law is applicable at constant pressure.
- ✓ Charles's law is not applicable at high pressure and low temperature.

"AVOGADRO'S LAW"**INTRODUCTION:**

YEAR: 1811

SCIENTIST: Amedeo Avogadro

STATEMENT:

"Equal volumes of all the gases contain equal number of molecules under the similar conditions of temperature and pressure."

OR

"At constant temperature and pressure, the volume of a gas is directly proportional to its number of molecules."

MATHEMATICALLY:

Let 'V' be the volume of a gas and 'n' be its number of molecules then according to Avogadro's law,

$$V \propto n \text{ (at constant temp and pressure)}$$

$$V = (\text{constant}) n$$

$$V = K n$$

Where,

V = volume of the gas.

K = constant of proportionality

n = number of moles.

APPLICATIONS:

- ✓ It helps to determine molecular mass of a gas provided its volume is known at S.T.P.
- ✓ It helps to determine volume of a gas provided its molecular mass is known at S.T.P.

ACCORDING TO K.M.T:

According to K.M.T, when two gases have same temperature, their K.E is same. So, they exert same pressure, because molecules collide with the walls with same force.

Thus, if gases have same temperature and volume and exert same pressure, it means they must have same number of molecules.

"GENERAL GAS EQUATION"**INTRODUCTION:**

There are three variables that are volume, pressure and temperature. These can be interrelated in an expression called the general gas equation. This equation can be derived by combining Boyle's law, Charles's law and Avogadro's law.

DERIVATION:

According to Boyle's law,

$$V \propto 1/p \text{ (at constant temperature)}$$

According to Charles's law,

$$V \propto T \text{ (at constant Pressure)}$$

According to Avogadro's law,

$$V \propto n \text{ (at constant P and T)}$$

By combining these three laws, we get:

$$V \propto nT / P$$

$$V = (\text{constant}) nT / P$$

$$V = RnT/P$$

Where "R" is the general gas constant. On further solution, we get:

$$\boxed{PV = nRT}$$

This equation is known as the general gas equation.

GENERAL FORM OF GENERAL GAS EQUATION

For 1 mole of gas, the general gas equation becomes

$$PV = RT$$

OR,

$$PV / T = R$$

For initial state:

$$P_1 V_1 / T_1 = R \dots \rightarrow (1)$$

For final state:

$$P_2 V_2 / T_2 = R \dots \rightarrow (2)$$

By comparing equation (1) & (2)

$$P_1 V_1 / T_1 = P_2 V_2 / T_2$$

This relationship is used to solve problems regarding changes of volume of gases, due to the changes in the pressure & temperature.

"VALUES & UNITS OF R"

In the general gas equation, the numerical value of universal gas constant (R) depends upon the units of pressure, volume and temperature.

IN CHEMICAL SYSTEM:

According to Avogadro's law, at S.T.P the one mole of any gas occupies a volume of 22.4 dm³ i.e.

$$T = 0^\circ\text{C} = 273^\circ\text{K}$$

$$P = 1 \text{ atm.}$$

$$n = 1 \text{ mole}$$

$$V = 22.4 \text{ dm}^3$$

Since,

$$PV = nRT$$

Therefore,

$$R = \frac{PV}{nT}$$

Putting the values, we get

$$R = \frac{(1 \text{ atm})(22.4 \text{ dm}^3)}{(1 \text{ mole})(273 \text{ K})}$$

$$R = 0.0821 \text{ atm. dm}^3 \cdot \text{Mol}^{-1} \cdot \text{K}^{-1}$$

IN S.I UNIT:

When 'P' is expressed in N/m² and volume 'V' in m³, then at S.T.P,

$$T = 0^\circ\text{C} = 273^\circ\text{K}$$

$$P = 101300 \text{ N/m}^2$$

$$n = 1 \text{ mole}$$

$$V = 0.00224 \text{ m}^3$$

Since,

$$PV = nRT$$

Therefore,

$$R = \frac{PV}{nT}$$

Putting the values, we get

$$R = \frac{\left(\frac{101300 \text{ N}}{\text{m}^2}\right)(0.00224 \text{ m}^3)}{(1 \text{ mole})(273 \text{ K})}$$

$$R = 8.314 \text{ N.m}^2 / \text{mole.K}$$

Since, $\text{N.m}^2 = \text{Joule}$

$$R = 8.314 \text{ J/mole.K}$$

"GRAHAM'S LAW OF DIFFUSION"

INTRODUCTION:

YEAR: 1881

SCIENTIST: Graham

STATEMENT:

"At constant temperature and pressure, the rate of diffusion of any gas is inversely proportional to the square root of its density."

	Ar	O ₂	N ₂
Volume:	22.4 L	22.4 L	22.4 L
Mass:	40 g	32 g	28 g
Quantity:	1 mol	1 mol	1 mol
Pressure:	1 atm	1 atm	1 atm
Temperature:	273 K	273 K	273 K

MATHEMATICALLY:

Mathematically, graham's law can be expressed as:

$$r \propto \frac{1}{\sqrt{d}} \quad \text{OR} \quad r = K \frac{1}{\sqrt{d}}$$

Where, r = rate of diffusion of gas,

d = density of gas,

K = Proportionality constant.

Suppose two gases with densities d_1 & d_2 , diffuse into each other. If the rate of diffusion of the gases are r_1 & r_2 respectively, then according to graham's law:

For gas 1,

$$r_1 = K \frac{1}{\sqrt{d_1}} \quad \text{----- i}$$

For gas 2,

$$r_2 = K \frac{1}{\sqrt{d_2}} \quad \text{----- ii}$$

By dividing equation i by ii, we get

$$\frac{r_1}{r_2} = \frac{K \frac{1}{\sqrt{d_1}}}{K \frac{1}{\sqrt{d_2}}}$$

OR

$$\frac{r_1}{r_2} = \frac{\sqrt{d_2}}{\sqrt{d_1}} \quad \text{-----} \rightarrow (A)$$

Since,

$$d = M/V$$

So, equation A becomes,

$$\frac{r_1}{r_2} = \frac{\sqrt{\frac{M_2}{V}}}{\sqrt{\frac{M_1}{V}}}$$

Since volume is constant so,

$$\frac{r_1}{r_2} = \frac{\sqrt{M_2}}{\sqrt{M_1}}$$

OR

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} \dots \rightarrow (B)$$

Since,

$$r = \frac{1}{t}$$

Therefore, equation (B) becomes,

$$\frac{\frac{1}{t_1}}{\frac{1}{t_2}} = \sqrt{\frac{M_2}{M_1}}$$

$$\frac{t_2}{t_1} = \sqrt{\frac{M_2}{M_1}}$$

K.M.T EXPLANATION:

According to K.M.T, The average kinetic energy of gas molecules depends upon absolute temperature. There is a large distance between the gas molecules. When gases are mixed, the molecules of each gas move into the free spaces of other gas molecules. This is diffusion of gases. Now A/C to K.M.T, when the temperature of gases is same, their kinetic energy is also same.

$$\frac{1}{2}m_1v_1^2 = \frac{1}{2}m_2v_2^2$$

$$m_1v_1^2 = m_2v_2^2$$

$$\left(\frac{v_1}{v_2}\right)^2 = \frac{m_2}{m_1}$$

Taking root on B.S

$$\frac{v_1}{v_2} = \sqrt{\frac{m_2}{m_1}} \dots \dots i$$

Since, the ratio of the velocities is same as that of rate of diffusion. Therefore,

$$\frac{v_1}{v_2} = \frac{r_1}{r_2}$$

Putting in equation I, we get

$$\frac{r_1}{r_2} = \sqrt{\frac{m_2}{m_1}}$$

This is the equation of Graham's Law of Diffusion.

APPLICATIONS:

With the help of Graham's Law, we can determine:

- ✓ Rate of diffusion of different gases.
- ✓ Molecular mass of unknown gases.

"DALTON'S LAW OF PARTIAL PRESSURE"

INTRODUCTION:

YEAR: 1801

SCIENTIST: John Dalton

STATEMENT:

"At constant temperature and volume the total pressure of the gaseous mixture is always equal to the sum of partial pressures of all the gases present in the mixture."

MATHEMATICALLY:

In a mixture of gases A, B and C:

P_A = partial pressure of Gas A

P_B = partial pressure of Gas B

P_C = partial pressure of gas C

P_T = partial pressure of mixture

According to Dalton's Law of partial pressure,

$$P_T = P_A + P_B + P_C$$

DERIVATION:

Consider a mixture of gases A, B and C.

Let:

No. of moles of gas A = n_A

No. of moles of gas B = n_B

No. of moles of gas C = n_C

Therefore,

$$n_T = n_A + n_B + n_C \text{----- i}$$

According to General Gas Equation,

$$PV = nRT$$

$$P = \frac{nRT}{V}$$

For Gas A:

$$P_A = \frac{n_A RT}{V}$$

For gas B:

$$P_B = \frac{n_B RT}{V}$$

For gas C:

$$P_C = \frac{n_C RT}{V}$$

Since,

$$P_T = P_A + P_B + P_C$$

$$P_T = \frac{n_A RT}{V} + \frac{n_B RT}{V} + \frac{n_C RT}{V}$$

$$P_T = \frac{RT}{V}(n_A + n_B + n_C)$$

$$P_T = \frac{RT}{V}(n_T)$$

Dividing P_A and P_T we get,

$$\frac{P_A}{P_T} = \frac{n_A RT}{V} \div \frac{RT}{V}(n_T)$$

$$\frac{P_A}{P_T} = \frac{n_A}{n_T}$$

Similarly,

$$\frac{P_B}{P_T} = \frac{n_B}{n_T}$$

And,

$$\frac{P_C}{P_T} = \frac{n_C}{n_T}$$

Also we can write as,

Partial pressure of a gas	=	moles of the gas
Total pressure		Total moles

I.e. the ratio of partial pressure of a gas to the total pressure of the mixture is equal to the mole fraction.

K.M.T EXPLANATION:

According to K.M.T, "There is no attraction between gas molecules". Hence, in the mixture of gases, each gas behaves independently. Molecules of each gas on colliding with the walls of container produce their own pressure. Hence the total pressure of the gaseous mixture is the sum of partial pressure of individual gas.

LIMITATIONS OF LAW:

This law is only applicable for gases which do not react with each other.

"APPLICATION"

COLLECTION OF GASES OVER WATER:

Dalton's law of partial pressure is useful for calculating volume of gas collected over water. When a gas is collected over water, some vapors are also present in the gas. This is called "wet gas" and its pressure is equal to the sum of pressure of dry gas and water vapour pressure therefore according to Dalton's law,

$$P_{\text{(wet gas)}} = P_{\text{(dry gas)}} + P_{\text{(vapour)}}$$

The partial pressure of water vapours is also called aquoustention,

$P_{\text{(dry gas)}} = P_{\text{(wet gas)}} - P_{\text{(aquoustention)}}$
--

"IDEAL GAS"

DEFINITION:

The gas which obeys all the gas laws, of all temperature & pressure and its behavior can be explained on the basis of Kinetic molecular theory is called IDEAL GAS."

PROPERTIES:

The ideal gas has the following two significant properties.

- ✓ The molecules of the ideal gas don't attract to each other.
- ✓ The size of the molecules is negligible or they have not space to occupy.

In actual practice, there exists no ideal gas in the nature, but the gases like H_2 or He show ideal behavior at specific temperature and pressure.

"DIFFERENCES"

<u>IDEAL GAS</u>	<u>REAL GAS</u>
Gas which obeys all gas law at all temperatures and pressures is called ideal gas.	Gas which obeys all gas laws at a certain temperature and pressure is called a real gas.
The size of molecules is zero.	The size of molecules is not zero.
No force of attraction or repulsion is present between the molecules.	They have weak attractive force between the molecules which is called Vander wall force.
This is an imaginary gas which does not exist practically.	They are real gases and exist practically.

"CAUSES OF DEVIATION"

The real gases deviate from ideal behavior due to the following faulty assumptions of the Kinetic Theory.

✓ VOLUME OF MOLECULES:

The total volume of the gas particles is negligible as compared to the volume of the gas itself.

EXPLANATION:

This may be acceptable at low pressure but at high pressure the molecules are forced together, as a result, the volume of a gas decreases while the total volume of its molecules

remains unchanged. Thus the total volume of its molecules cannot neglect as compared to the volume of gas itself.

✓ INTERMOLECULAR FORCES:

There are no attractive or repulsive forces between the gas particles.

EXPLANATION:

This is also not true at high pressure because when the pressure is increased, the gas molecules become closer, and thus attractive forces increase. This attractive force between the molecules is known as Inter-molecular forces. These intermolecular forces cause the real gas to deviate from ideal behavior.

VAN DER WAAL'S EQUATION

In order to modify the two conflicting postulates of the Kinetic Molecular theory regarding volumes and intermolecular attraction of gas molecules, a Dutch scientist, J.D. van der Waal (1873), gave a mathematical solution by the correction of molecular volume and intermolecular forces. This is known as the van der Waal equation.

VOLUME CORRECTION:

When pressure is applied to a gas, the molecules come closer to each other. By continuous increase in pressure, a point is reached when molecules cannot further be compressed since repulsive forces are created. This indicates that gas molecules have a definite volume. Although this is very small but not negligible. Keeping in view the definite volume of gas molecules, van der Waal calculated the actual volume of a gas as follows:

$$V = V_{\text{vessel}} - nb \text{ -----(i)}$$

Where,

V = free volume

V_{vessel} = Volume of the vessel in which gas molecules are present.

n = number of moles

b = excluded volume of gas molecules per mole in the highly compressed gaseous state. It is theoretical volume.

The value of ' b ' depends upon the size of gas molecules. Here it should be noted that ' b ' is not actual volume of a gas but it is assumed four times the actual volume of molecules.

Where, V_{molecule} is actual volume of one mole of a gas.

PRESSURE CORRECTION:

Van der Waal also corrected the pressure produced by molecules of the real gases. As we have already discussed that intermolecular attraction comes into play when the molecules are brought close together by squeezing the gas. Consider a molecule A in the interior of the gas, which is completely surrounded and attracted by the other gas molecules B. The resultant attractive force on molecule A due to all the surrounding B molecules is zero as shown in figure 4.5b (in book). However, as this molecule A approaches to strike the wall of the vessel, it experiences a net inward pull due to the attractive forces of molecule of B. It means that the pressure produced on the wall would be a little bit lesser than the pressure of an ideal gas molecule.

Therefore,

$$P_{\text{observed}} = P_{\text{ideal}} - P_{\text{less}}$$

If observed pressure is simply indicated by P, ideal pressure P_i , and less pressure P_L , Then equation will be:

$$P = P_i - P_L$$

OR

$$P_i = P + P_L \text{----- (ii)}$$

Since P_L is the pressure drop due to backward pull of striking molecules, it depends upon number of particles (A and B) per unit volume.

$$P_L \propto [A] \cdot [B]$$

$$P_L \propto \frac{n^2}{v^2}$$

$$P_L = \frac{an^2}{v^2}$$

Insert the value of P_L in equation (ii).

$$P_i = P + \frac{an^2}{v^2} \text{----- (iii)}$$

The corrected pressure and volume is now put in general gas equation, we get.

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

If $n=1$ then equation will be:

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT \dots\dots\dots (V)$$

This is van der Waal's equation. Here 'a' and 'b' are van der Waal's constants.

LIQUEFACTION OF GASES

Liquefaction is a physical process or technique in which gases are converted into liquids. Gas molecules at high pressure and low temperature come closer and attractive forces start operating. When the temperature is further lowered the attractive forces draw the molecules together to form a liquid.

Generally, it has been observed that, every gas has a specific temperature above which gas cannot be converted into liquid even at high pressure. The highest temperature, at which a gas can exist as a liquid, is called its **Critical Temperature**. While the pressure which is required at critical temperature to convert a gas into liquid is known as **Critical Pressure**. However, critical volume depends upon critical temperature and pressure. At critical the volume of one mole of a gas is called **Critical Volume**.

JOULES-THOMPSON EFFECT:

When a compressed gas is allowed to expand from a region of higher pressure to a lower pressure through a nozzle causes a fall in temperature. This is known as Joule Thomson effect. The fall in temperature is proportional to the pressure difference between compressed and expanded gas. The molecule of gas in low pressure area absorb energy from environment to reduce their attractive forces. The most common example of this effect is applied to liquefy Nitrogen and carbon dioxide (dry ice).

"LINDE'S METHOD"

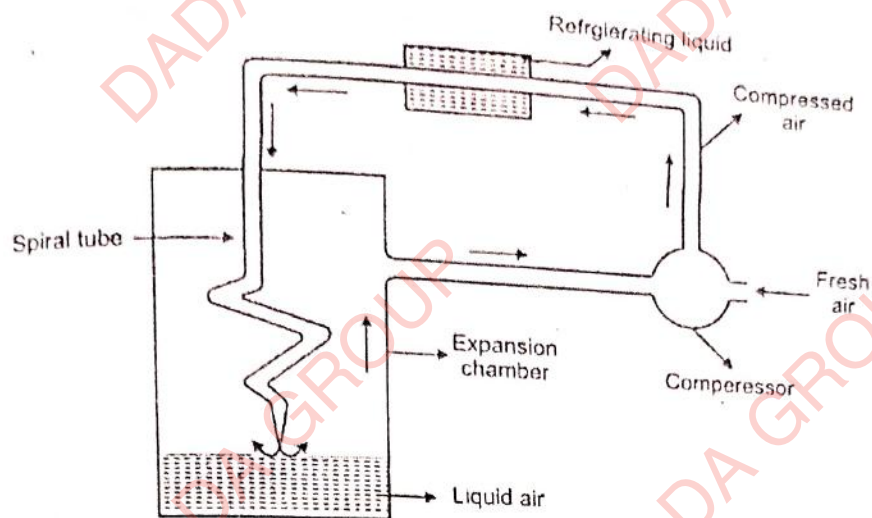
INTRODUCTION:

Hampson-Linde (1842-1934) introduced this method based on the principle of the Joule-Thomson effect for the liquefaction of gases.

STATEMENT:

"When a gas is allowed expand in an adiabatic closed system from a region of higher pressure to a region of extremely lower pressure, the liquefaction of air or any gas takes place."

CONSTRUCTION:



WORKING:

In this method, first of all, fresh air or any other gas is compressed containing water. Here moisture present in air is condensed as water and removed. In this part, the atmosphere in a compressor. When compressed air is allowed to pass through pipe where the heat of compressed air is absorbed by the water and dry air moves through the copper spiral coil having a jet at the end. When dry air passes through it, expansion occurs pressure is reduced to one atmosphere. This causes a decrease in the temperature of the air. As a result, cooled air is formed in the expansion chamber which moves up and cools the incoming compressed air. Then uncondensed air from the expansion chamber returns to the compression pump for recycling. This process of cooling and compression followed by expansion is repeated many times. Thus, liquid air is collected at the bottom of the expansion chamber and removed.

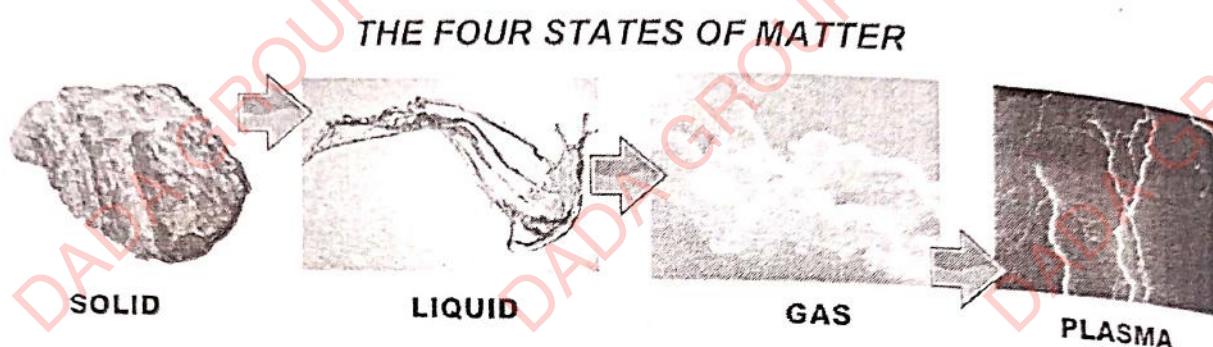
LIQUID AIR AND ITS USES:

Ordinary air on compression and cooling to extremely low temperature becomes liquefied and known as liquid air. It is mainly used for the cooling purpose. It absorbs the heat very rapidly and helps to convert the gases (filled in containers) into liquids and liquids into the solids. It is the industrial source of various gases such as oxygen, argon, nitrogen etc through a process named as air separation. These liquefied gases have various applications in various fields and common life. Some important uses of liquefied gases are given below:

- i. Liquefied natural gas (L.N.G) is used as a fuel.
- ii. Liquefied petroleum gas (L.P.G) is a mixture of butane and other hydrocarbons, used as fuel for portable heaters, gas cooktops and ovens. It is also used as fuel for engines.
- iii. Ammonia and liquid sulphur dioxide are used as refrigerants.
- iv. Liquid air is an important source of oxygen in rockets.
- v. Compressed oxygen is used for welding purpose.
- vi. Nitrogen is used in medical science.

FOURTH STATE OF MATTER: PLASMA

Except solid, liquid and gas, the fourth state of matter is called "Plasma". This word is taken from the Greek "plassein" which means moldable substance. This state of matter was identified by William Crooks in 1879 while he was working on "Discharge Tube Experiment". He called it "radiant matter". But later on, in the 1920s, Irving Langmuir used the word Plasma". Before solids, liquids and gases, plasma was existed in the universe. More than 99% of the visible universe is composed of plasma. Plasma occurs naturally in the sun, stars, nebulae and ionosphere, but rarely on our Earth. The shining of stars and the sun is due to the plasma. On Earth it is found in lightning bolts, flames, welding arcs, auroras and fluorescent lights.



When a gas is heated at high temperature (10^4 to 10^5 K), most of the molecules are ionized along with formation of a cloud of electrons. Despite high temperature, there still exist some un-ionized molecules and atoms.

PLASMA:

"The mixture of positive ions, electrons and un-ionized molecules and atoms is known as a plasma".

EXISTENCE:

Natural plasma exists only at high temperature or low temperature vacuum. Natural plasma does not break down or react rapidly but it is extremely hot (over 20000°C).

ENERGY:

Its energy is so high that it can vaporize any matter.

NEUTRAL ATOM:

Plasma as a whole is neutral because it contains equal number of negatively charged electrons and positively charged ions. Plasma contains a significant number of charged particles which affect its properties. The presence of charged particles enables it to respond to electrical and magnetic fields.

SHAPE:

Like gases, plasma has neither definite shape nor volume.

GLOW CHARACTERISTIC:

Plasma shows characteristic glow in the discharge tube depending upon the nature of the gas. For example, hydrogen exhibits green, oxygen red and nitrogen purple glows.

USES OF PLASMA:

Plasma has enormous uses in various fields which are given below:

- i. When light of a fluorescent bulb or neon sign is turned on, gas is excited and creates glowing plasma which lightens the surroundings
- ii. It is used in television and computer chips.
- iii. Plasma is used in rocket propulsion, cleaning the environment, destroying biological hazards, and healing wounds.
- iv. It is used in the processing of semiconductors and sterilization of some medical products.
- v. It is used in lasers, lamps, pulsed power switches, and diamond-coated films.

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